Class E inverter design [1]

For an arbitrary duty cycle, a finite input inductance, and arbitrary commutation conditions

I – Circuit diagram

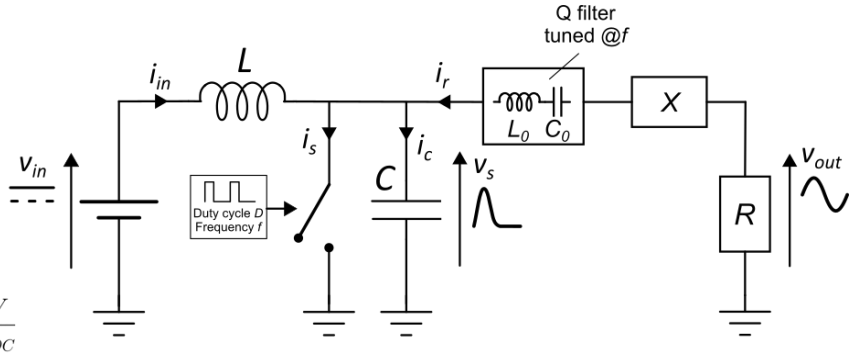
$$D = \frac{T_{ON}}{T}$$

$$q = \frac{1}{\omega\sqrt{LC}}$$

$$K = v_s(2\pi)$$

$$s = \frac{\mathrm{d}v_s(\theta = 2\pi)}{\mathrm{d}\theta}$$

$$R_{DC} = \frac{< v_{in} >}{< i_{in} >} = \frac{V}{I_{DC}}$$



II – Design assumptions

- Transistor is ideal (no power losses and instantaneous switching)
- Passive components are ideal (no power losses)
- Capacitance C is linear (constant)
- Current i_r is sinusoidal at the switching frequency f (Q high enough)

III – Structure of the MATLAB script

To be downloaded at ***https://github.com/matthieubeley/class-E-design***

1) Inputs

Inputs are the duty cycle D and the q ratio. Other inputs are the switch voltage K and the switch voltage derivative s at turn-on. For nominal class E commutation conditions (ZVS and ZdVS), they are both equal to 0.

These inputs are either scalars (for investigating a single point) or vectors (for the complete exploration of the design space).

*2) Parameters calculation (for the **normalized** circuit)*

Analytical calculations only. All normalized circuit parameters are obtained.

3) De-normalization

Following class E parameters are obtained, considering the actual operating point (input DC voltage V, switching frequency f and load resistance R).

Circuit parameters	**Voltages**	**Currents**
Input inductance L	Output voltage mag. V_{out}	Input DC current I_{in}
Shunt capacitance C	Switch voltage fund. V_{s1}	Output current magnitude I_r
Residual reactance X	Peak switch voltage V_{sp}	Peak switch current I_{sp}
Output power P		Switch RMS current $I_{s_{RMS}}$
Equivalent DC input resistance R_{DC}		

Contour plots for parameter ***x*** (isolines ***x***=a, ***x***=b, etc.) can be displayed using function *contour3(q,D,**x**,[a,b,…])*. This is how the different charts in the shared folder have been obtained.

4) Waveforms

ONLY ONCE the inputs q, D, K and s are set scalars!
Main circuit voltages and currents temporal waveforms can be plotted.

Voltages	**Currents**
Switch voltage v_s	Output current i_r
Switch voltage fundamental v_{s_1}	Input current i_{in}
Output voltage v_{out}	Switch current i_s
	Shunt capacitor current i_c

[1] M. Beley, L. Pace and A. Bréard, "Performances Assessment of Very High-Frequency Class E Inverters Based on a Load-Oriented Generic Design Method," in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 13, no. 4, pp. 4721-4733, Aug. 2025, doi: 10.1109/JESTPE.2025.3565020.